BRIGHT STARS:

TECHNOLOGY-MEDIATED SETTINGS FOR URBAN YOUTH AS PATHWAYS FOR ENGAGED LEARNING

CEHD
COLLEGE OF EDUCATION + HUMAN DEVELOPMENT
UNIVERSITY OF MINNESOTA

Cynthia Lewis, Ph.D. | Cassandra Scharber, Ph.D. | Tracey Pyscher | Kris Isaacson

BACKGROUND

This study examines the complex features of high-quality, technology-rich school, library, and community-based settings that position urban youth as knowledge producers and meaning-makers through the use of technologies. Our purpose is to determine features within and across technology-rich settings that promote and result in engagement-in-learning to inform both within and cross-setting practices, fuel potential collaborations, and create better learning environments for urban youth (Ito, et al., 2013).

How do school, library, and community-based urban settings position youth as engaged learners through the use of digital technologies?

1. What are the features of each setting that promote engaged learning?
2. What is the nature of learning in each setting?
3. How does the activity system influence engagement-in-learning in each setting?

THEORETICAL FRAMEWORKS

Connected Learning

The increasing emphasis on "transforming education" using technology-infused learning in national policy initiatives (USDOE, 2010; H.R.521, 2013) underscores the importance of moving beyond technology skill acquisition to "connected learning" (Ito et al., 2013). Connected learning focuses on learning pathways that move across formal and informal settings that transform the very nature of learning – what it means, how it occurs, and where it takes place (Barron & Hammond, 2010; Hobbs & Frost, 2003; Jenkins et al., 2009; Peppler, 2013).

Activity Theory

Within Activity Theory (AT), activity is goal-oriented and mediated by symbolic and concrete tools that have social, cultural, and historical uses within the system (Engeström, 1999). Learning in activity systems occurs between individuals and the people, tools, and symbols within the system; therefore, individuals are indivisible from their social contexts. The elements of activity systems interact in ways that must be understood in order to arrive at a nuanced understanding of engagement as a dynamic process with links to learning.

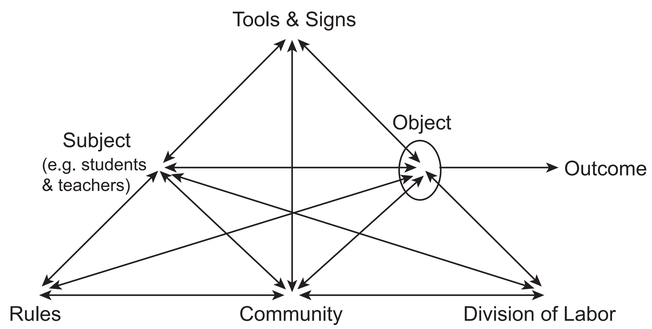


Figure 1

Engagement in Learning

Engagement is conceived of as multidimensional, with interrelated behavioral, emotional, and cognitive aspects (Barko, 2005). Learning is defined as a social practice dependent upon interaction among people and their tools and technologies (Engeström, 2010; Lave, 1996; Rogoff, 1995; Wenger, 1999). Engagement-in-learning is understood through a lens of Activity Theory, which honors the complexity of engagement in different types of learning contexts.

RESEARCH METHODS

Settings and Participants

This study focused on 3 formal and non-formal learning sites within the Twin Cities where youth (ages 12-18) actively used and created with digital media through consistent and sustained access to technology. The focus was on 1 school classroom (S), 1 library program (L), and 1 community-based organization (C). Participants included youth and facilitators at each site.



Site Selection

In order to identify high-quality technology-mediated sites, we used criteria grounded in both practice and theory. These *practice-driven* criteria guided our inclusion and selection of research sites:

- Youth actively use and create with digital media;
- Youth have consistent and sustained access to technology;
- Digital media production plays a central role in the learning processes;
- Recognition by external community or other entities as having quality programs (grants, awards).

Research sites also met Jenkins' (2009) *theoretically-based* criteria for supporting "participatory culture:"

- with relatively low barriers to artistic expression and civic engagement;
- with strong support for creating and sharing one's creations with others;
- with some type of informal mentorship whereby what is known by the most experienced is passed along to novices;
- where members believe that their contributions matter;
- where members feel some degree of social connection with one another.

Design, Methods, Analysis

We utilized a multi-case study design (Yin, 2009) with qualitative methods to study school, library, and community-based learning settings. First, case descriptions were composed for each setting. Next, a cross-case analysis was conducted that examined features across all three settings (S, L, C).

Observations

Observations were our primary data source and provided us with insights into the features of each activity system, as well as learning and engagement in each setting. Two methods were used to collect data during observations: *Qualitative fieldnotes* and a formal protocol, the *Activity System Observation Protocol (ASOP)*. During each site visit, researchers in the role of participant-observer took fieldnotes that focused primarily on description with some supplementary interpretive comments. The ASOP provided a detailed account of the components of a learning activity as enacted in a given session or class period.



Interviews

The student/youth interviews focused on understanding how digital tools and other components of the activity system affected their engagement and learning. The teacher/facilitator interviews focused on understanding her/his perceptions of learning and engagement within the setting and the impact of pedagogical practices and technology.

Artifacts and Documents

Artifacts and documents including mission statements, rules, expectations, guidelines, curriculum materials, and youth-created projects/products were collected for describing the features of each technology-mediated activity system.

Data Summary

Classroom: October – March 2012
27 observations; 20 students & 1 teacher; 7 interviews; technology-use survey; 140 pages of field notes/ASOP

Community-Based Organization: June – July 2013
14 observations; 12 youth, 2 youth mentors, 2 adult facilitators; 7 interviews; 137 pages of field notes/ASOP

Library: December 2012 – May 2013
24 observations; 3 youth & 1 adult facilitator; 4 interviews; technology-use survey; 145 pages of field notes/ASOP

FINDINGS

Objects Differed Across Settings:

- **Classroom:** using texts and technologies for identity affiliation and social critique
- **CBO:** community engagement and developing expertise
- **Library:** developing expertise through mentorship for equity purposes

Class (English in an Urban High School): Two Features of Engagement

Media-Saturated Intertextuality and Critique: Whole class and peer discussions frequently included impromptu uses of technology to enhance learning. These uses of digital technologies quickly moved from classroom texts and talk to media texts and sources, drawing connections in the service of social commentary and social justice.

Dialogic Talk: Characterized by student -generated questions, seamless weaving of formal and informal language, official and unofficial knowledge, "new" (digital medial) and "old" (print texts) technologies, and animated sharing of multiple perspectives that continually referenced previous speakers, class texts, and media sources.

"I mean I don't feel like I am actually getting something put in my head . . . In here [English class], it's like we make up what's right and we create . . . our own work to learn from others and ourselves" (Jenda)

Community-Based Organization: "Fun and Challenging" Engaged Learning

The CBO's OBJECT was taken up intently, enthusiastically, and progressively by youth through collaborative, mentored experiences: 1) media production skills, 2) work readiness & career consideration, 3) community exploration. Youths' talk makes visible the progression and construction of object realization. Also, it is through language that adults mentored/scaffolded the youth into roles of documentary film producers, employed teens, creative artists, community members, and technology technicians.

"We are done! We finished the 1st question!" (Kaycee). Kaycee and Alim finish merging together video clips during intro editing session with Adobe Premiere Pro CS6. "We are like professionals" (Alim, big smile). [Week 2]

Library: Developing Expertise and Leadership through Learning

The youth were engaged in learning when they (1) felt at ease and comfortable with one another despite the diverse makeup of squad & workshop teens and library staff, (2) trusted and felt safe to freely experiment with technology, (3) when there was an equitable division of labor where the adults did not hold the exclusive role of expert, and (4) could control the direction of their learning based on their own interests.

"I can use what we do here in school. For example, iMovie. So, I use it here and I know how to make trailer and stuff. And, then for school, we do like, projects, iMovies and stuff. So, it's like, 'I know this. I can help you guys.'" (Mahdi)

Common Traits Across Settings:

The Object was not dictated but could, instead, be shaped by youth in each of the three settings. The facilitator expectations for the youth were, (1) to take up and transform the object, (2) to share in decision making and participation, and (3) to bring their own related purposes to bear.



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